45 YEARS OF EXCELLENCE | NARAYANA IIT-JEE ACADEMY-INDIA

JR IIT CO SUPER CHAINA **Dt : 01/07/2024**

Time: 3HOURS **CAT-01** **Max.Marks:198**

JEE-ADVANCE-2020-P2-Model

IMPORTANT INSTRUCTIONS

PHYSICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 1 – 6)	Questions with Single digit integer(0-9)	+3	-1	6	18
Sec – II (Q.N : 7 – 12)	One of More Correct Options Type (partial marking scheme) (+1)	+4	-2	6	24
Sec – III (Q.N : 13 – 18)	Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127. 30)	+4	0	6	24
Total				18	66

CHEMISTRY

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 19 – 24)	Questions with Single digit integer(0-9)	+3	-1	6	18
Sec – II (Q.N : 25 – 30)	One of More Correct Options Type (partial marking scheme) (+1)	+4	-2	6	24
Sec – III (Q.N : 31 – 36)	Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127 .30)	+4	0	6	24
Total				18	66

MATHEMATICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 37 – 42)	Questions with Single digit integer(0-9)	+3	-1	6	18
Sec – II (Q.N : 43 – 48)	One of More Correct Options Type (partial marking scheme) (+1)	+4	-2	6	24
Sec – III (Q.N : 49 – 54)	Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127 .30)	+4	0	6	24
Total				18	66

PHYSICS **Max.Marks:66**

SECTION – 1 (Maximum Marks: 18)

This section contains **SIX (06)** questions.
The answer to each question is **A SINGLE DIGIT INTEGER** ranging from **0 TO 9 , BOTH INCLUSIVE**.
For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If only the correct option is chosen.
Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)
Negative Marks: -1 In all other cases.

1. A bird moves with velocity of $20\,ms^{-1}$ in the direction making angle 60^0 with eastern line and 60^0 with vertically upward. Represent the velocity vector in rectangular form is $\bar{v} = x\hat{i} + y\sqrt{z}\,\hat{j} + w\hat{k}$ then find the value of $\frac{xz}{y}$

2. Let $\vec{a},\vec{b}$ and $\vec{c}$ be three vectors such that $\vec{a}+\vec{b}+\vec{c}=\vec{0}$. If $|\vec{a}|=3,\ |\vec{b}|=4$ and $|\vec{c}|=5$, then find the value of $\left|\left(\vec{a}.\vec{b}+\vec{b}.\vec{c}+\vec{c}.\vec{a}\right)\right|$ is 5x then find the value of x

3. Given: $|\vec{A}_1|=2, |\vec{A}_2|=3$ and $|\vec{A}_1+\vec{A}_2|=3$. Find the value of $\left|\left(\vec{A}_1+2\vec{A}_2\right).\left(3\vec{A}_1-4\vec{A}_2\right)\right|$ is x^2 then find the value of x

4. Six forces of magnitude 1,2,3,4,5 and 6 act along the sides of a regular hexagon taken in order. Resultant has magnitude

5. The x and y components of $\vec{A}$ are 4 m and 6m, respectively. The x and y components of $\left(\vec{A}+\vec{B}\right)$ are 10m and 9m respectively. The magnitude of vector $\vec{B}$ is $3\sqrt{x}$ Find x.

6. The area of a blot of ink is growing such that after t second, its area is given by $A=\left(2t^2+7\right)cm^2$. Calculate the rate of increase of area at t = 5 seconds is $x\times 10^1\left(C.G.S\right)$

SECTION - 2 (Maximum Marks : 24)

This section contains **SIX (06)** questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

7. Van der Waal's gas equation for n number of moles is given by the formula $\left(P+\frac{n^2a}{V^2}\right)(V-nb)=nRT$, where P is pressure, V is volume, R is gas constant and T is temperature

A) Dimensions of a is $ML^5T^{-2}\,mol^{-2}$
B) Dimensions of b is L^3mol^{-1}
C) Dimensions of R is $ML^2T^{-2}\,mol^{-1}K^{-1}$
D) Dimension of $\frac{a}{b}$ is $MLT^{-2}\,mol^{-1}$

8. The equation of a stationary wave is $y=2A\sin\left(\frac{2\pi ct}{\lambda}\right)\cos\left(\frac{2\pi X}{\lambda}\right)$ which of the following is correct ?

A) Unit of ct is same as that of λ
B) Unit of X is same as that of λ
C) Unit of $\frac{2\pi c}{\lambda}$ is same as that of $\frac{2\pi X}{\lambda t}$
D) Unit of $\frac{c}{\lambda}$ is same as that of $\frac{X}{\lambda}$

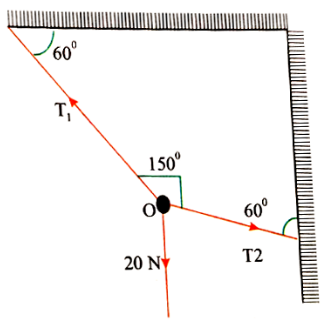
9 The energy 'E' of a particle varies with time t according to the equation $E=E_0\sin(\alpha t)e^{\frac{-\alpha t}{\beta x}}$; where x is displacement from mean position, E_0 is energy at infinite position and α & β are constants

A) Dimension of α is LT^{-1}
B) Dimension of β is $M^0L^{-1}T^0$
C) Dimension of $\sin\left[\frac{\alpha t}{\beta x}\right]$ is $M^0L^0T^0$
D) Dimension of α is T^{-1}

10 If the dimensions of length are expressed as $G^xc^yh^z$; where G, c and h are the universal gravitational constant, speed of light and Planck's constant respectively, then

A) $x=\frac{1}{2},y=\frac{1}{2}$
B) $x=\frac{1}{2},z=\frac{1}{2}$
C) $y=\frac{1}{2},z=\frac{3}{2}$
D) $y=\frac{-3}{2},z=\frac{1}{2}$

11 If 'O' is at equilibrium, then the values of the tensions T_1 and T_2 are (20 N is acting vertically downwards at O)



A) 20N, 30N
B) $20\sqrt{3}N,20\,N$
C) $20\sqrt{3}N,20\sqrt{3}\,N$
D) $10N,30\,N$

12. Dimensions of light year are the same as those of

A) leap year
B) wavelength
C) radius of gyration
D) propagation constant

SECTION - 3 (Maximum Marks : 24)

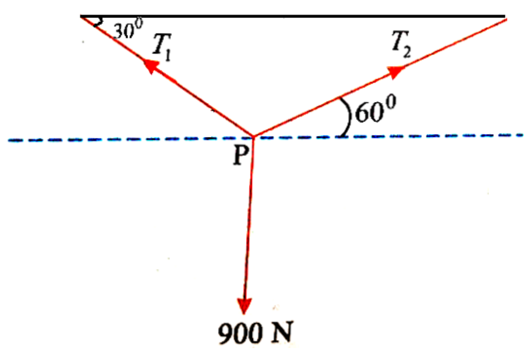
This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**
For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer. If the numerical value has more than two decimal places **truncate/round- off** the value to **TWO** decimal places.
Answer to each question will be evaluated **according to the following marking scheme:**
Full Marks: +4 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

13. If the time period t of the oscillation of a drop of liquid of density d, radius r, vibrating under surface tension s is given by the formula $t=\sqrt{r^{2b}s^{c}d^{a/2}}$. It is observed that the time period is directly proportional to $\sqrt{\frac{d}{s}}$. The value of b should be:

14 Two find the distance d over which a signal can be seen clearly in foggy conditions, a railways - engineer uses dimensions and assumes that the distance depends on the mass density ρ of the fog, intensity (power/ area) S of the light from the signal and its frequency f. The engineer finds that d is proportional to $S^{1/n}$. The value of n is

15 The square of the resultant of two forces 4N and 3N exceeds the square of the resultant of the two forces by 12 when they are mutually perpendicular. The angle between the vectors in degrees is

16. If 'P' is in equilibrium, then $\frac{T_1}{T_2}$ is nearly



17. Three forces $\vec{F}_1=\left(3\hat{i}+2\hat{j}-\hat{k}\right)N$, $\vec{F}_2=\left(3\hat{i}+4\hat{j}-5\hat{k}\right)N$ and $\vec{F}_3=A\left(\hat{i}+\hat{j}-\hat{k}\right)N$ act simultaneously on a particle. In order that the particle remains in equilibrium, the value of A should be

18 Side of a cube is increasing at the rate of 2 cm s^{-1}. Find the rate of increases of volume when the side length is 6 cm______ (CGS)

CHEMISTRY **Max.Marks:66**

SECTION – 1 (Maximum Marks: 18)

This section contains **SIX (06)** questions.
The answer to each question is **A SINGLE DIGIT INTEGER** ranging from **0 TO 9 , BOTH INCLUSIVE**.
For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If only the correct option is chosen.
Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)
Negative Marks: -1 In all other cases.

19. The number of neutrons present in radio active isotope of hydrogen is _______

20 The sum of the ratio of the radius difference between 4th and 3rd orbit of H-atom and that of Li^{2+} ion is _____

21. The work function (ϕ) of some metals are listed below. The number of metals which will show photo electric effect when light of 300 nm wave length falls on the metal is ___

Metal	Li	Na	K	Mg	Cu	Ag	Fe
$\phi(ev)$	2.4	2.3	2.2	3.7	4.8	4.7	4.3

22. A certain transition in hydrogen spectrum from an excited state to the ground state in one or more steps gives rise to total of '10' lines. How many of these belongs to UV- region ________

23. The frequency of a wave light is $1.0\times10^{6}\ \sec^{-1}$. The wave length for this wave is ______$\times10^{4}\ cm$

24. Calculate and compare the energies of two radiations one with a wave length of 300 nm and the other with 600 nm. Then $E_1 = x\times E_2$
The value of 'x' is ______

SECTION - 2 (Maximum Marks : 24)

This section contains **SIX (06)** questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

25. Many elements have non-integral atomic masses because

A) they have isotopes

B) their isotopes have non-integral masses

C) their isotopes have different masses

D) the constituents, neutrons, protons and electrons combine to give fractional masses

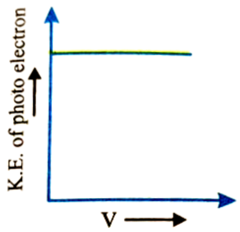
26 An isotone of ${}^{76}_{32}Ge$ is :

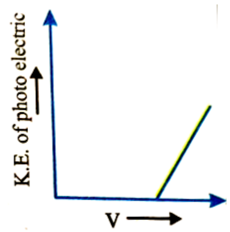
A) ${}^{77}_{32}Ge$ B) ${}^{77}_{33}Ge$ C) ${}^{77}_{34}Se$ D) ${}^{78}_{34}Se$

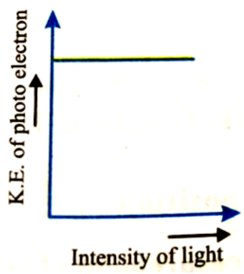
27. Which of the following have same significant figures ?

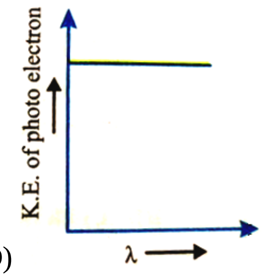
A) 6.02×10^{23} B) 7.70×10^{-20} C) 7.50 D) 0.75

28. Which is correct graph for photoelectric effect.

A) 

B) 

C) 

D) 

29 The energy of an electron in the first Bohr orbit of H atom is- 13.6eV. The possible energy value(s) of the excited state(s) for electrons in Bohr orbits of hydrogen is (are)

A) -3.4 eV B) -4.2 eV C) -6.8 eV D) -1.5 eV

30 Which statements concerning Bohr's model is/are true ?

A) Predicts that probability of electron near nucleus is more

B) Angular momentum of electron in H-atom = $\frac{nh}{2\pi}$

C) Introduces the idea of stationary states D) Explain line spectrum of hydrogen

SECTION - 3 (Maximum Marks : 24)

This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**
For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer. If the numerical value has more than two decimal places truncate/round- off the value to TWO decimal places.
Answer to each question will be evaluated **according to the following marking scheme:**
Full Marks: +4 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

31. The prefix 'Yotta' has value 10^{xy} here "$x\times y$" is ________

32. A measured temperature on Fahrenheit scale is 200^0F. What will this reading be on Celsius scale ______ C^0

33. One mole of photons, each of frequency $250\sec^{-1}$ would have approximately a total energy in ergs ______

34. How many times does the electron go round the first Bohr's Orbit of H in one second $P \times 10^{q}$ then $P+Q$ is ______

35. __________ $\times 10^{6}\, m/s$ velocity should an $\alpha -$ particle travel towords the nucleus of a copper atom to arrive at a distance of $10^{-13}\, m$ from the nucleus of the copper atom

36. The mass-charge ratio for $A^{\oplus}$ ion is $1.97 \times 10^{-7}\, Kg/C$, the mass of 'A' atom is __________ $10^{-26}\, Kg$

MATHEMATICS **Max.Marks:66**

SECTION – 1 (Maximum Marks: 18)

This section contains **SIX (06)** questions.
The answer to each question is **A SINGLE DIGIT INTEGER** ranging from **0 TO 9 , BOTH INCLUSIVE**.
For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If only the correct option is chosen.
Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)
Negative Marks: -1 In all other cases.

37. If mid points of sides BC, CA, AB of triangle ABC are (-1, 3) ; (-2, 4) ; (2, -5) respectively then length of median through vertex 'A' is $\sqrt{lm}$ then $l+m=$ ___

38. If $\cos\theta + \cos^{2}\theta = 1$ and $a\sin^{12}\theta + b\sin^{10}\theta + c\sin^{8}\theta + d\sin^{6}\theta = 1$

$$\Rightarrow \frac{b+c}{a+d} = \_\_\_\_$$

39. If $\sum_{i=1}^{n} Sin\theta_i = n$ then $\cos\theta_1 + \cos\theta_2 + \cos\theta_3 + ... + \cos\theta_n =$ _____________

40. $y = (0.36)^{\log_{0.25}\left(\frac{1}{3}+\frac{1}{3^2}+\frac{1}{3^3}+.....upto\,\infty\right)}$ then $10y =$ _____

41. $A=(3x_1, 3y_1); B=(3x_2, 3y_2); C=(3x_3, 3y_3)$ are vertices of a triangle with orthocentre 'H' at $(x_1+x_2+x_3\, ; y_1+y_2+y_3)$ then $\underline{|ABC} = \frac{\pi}{K}$ then $K =$ _______

42. $A = \{x : x = n^2 ; n = 1, 2, 3\}$ then number of proper subsets of A = _____

SECTION - 2 (Maximum Marks : 24)

This section contains **SIX (06)** questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

43. If $\frac{2x}{2x^2+5x+2} > \frac{1}{x+1}$ then $x \in$

A) $(-2,-1)$ B) $\left(\frac{-2}{3},\frac{-1}{2}\right)$ C) $(-2,-1)\cup\left(\frac{-2}{3},\frac{-1}{2}\right)$ D) R

44. If $4\{x\} = x+[x]$, $(\{\ \}\text{(fractionalpart) and [.] G.I.F})$ then x = _____

A) 0 B) $\frac{5}{3}$ C) $\frac{10}{3}$ D) 1

45. Solve the equation $\log_5\left(5^{\frac{1}{x}}+125\right) = \log_5 6 + \left(1+\frac{1}{2x}\right)$

A) $\frac{1}{2}$ B) $\frac{1}{4}$ C) $\frac{1}{5}$ D) 1

46. If $\frac{Sin^4 x}{2}+\frac{Cos^4 x}{3}=\frac{1}{5}$ then

A) $Tan^2 x = \frac{2}{3}$ B) $\frac{Sin^8 x}{8}+\frac{Cos^8 x}{27}=\frac{1}{125}$

C) $Tan^2 x = \frac{1}{3}$ D) $\frac{Sin^8 x}{8}+\frac{Cos^8 x}{27}=\frac{2}{125}$

47. If $P(1,2);Q(4,6);R(5,7);S(a,b)$ are vertices of parallelogram PQRS then

A) $a=2$ B) $b=3$ C) $a+b=5$ D) $ab=6$

48. If -1 be a zero of polynomial $x^3+6x^2+11x+6=0$ then other zeros of polynomial are

A) -2 B) -3 C) 2 D) 3

SECTION - 3 (Maximum Marks : 24)

This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**

For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer. If the numerical value has more than two decimal places truncate/round- off the value to TWO decimal places.

Answer to each question will be evaluated **according to the following marking scheme:**

Full Marks: +4 If ONLY the correct numerical value is entered as answer.

Zero Marks: 0 In all other cases.

49. The area of the pentagon whose vertices are (4, 1) ; (3, 6) ; (-5, 1) (-3, -3) and (-3, 0) respectively is_____

50 The polynomial equation whose roots are $2+3i, 2-3i$, $1+i;1-i$ is $x^4+bx^3+cx^2+dx+e=0$ then $b+c+d+e=$

51. The factorization of polynomial $P(x)=(x^2+x+1)(x^2+x+2)-12$ is $(x^2+x+d_1)(x+d_2)(x+d_3)$ then $d_1+d_2+d_3=$

52. The roots of the equation $4x^3-13x^2-13x+4=0$ are α,β,γ such that $\alpha<\beta<\gamma$ then $\alpha+4\beta+\gamma=$

53. The 10th common term between series $3+7+11+.....$ and $1+6+11+.......$

54. The degree of expression $(1+x)(1+x^6)(1+x^{11}).....(1+x^{101})$ is _________

PAPER SETTER:: NLR-NAIIT